

Football Scouting Intelligence System:

Player Similarity and Archetype Analysis

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Portfolio Case Study | StatsBomb Open Data | github.com/rahmahh53

Abstract

The system converts discrete on-ball events into standardized player profiles, uses Euclidean distance to identify stylistic neighbors, and applies K-Means to discover broad playing archetypes without position labels. The result is a transparent first-pass scouting workflow: define a target style, generate a shortlist, and send candidates to contextual and video review.

Research question

How can event data identify players with similar playing styles?

4,160

PLAYERS ANALYZED

13

MODEL FEATURES

6

PLAYING ARCHETYPES

5+

MATCHES REQUIRED

1. System outputs and data architecture

Outputs. Standardized Euclidean distance ranks stylistically similar players; K-Means separates the pool into interpretable archetypes. SQL-derived features keep the recommendations traceable, while the model remains a shortlisting aid rather than a replacement for contextual and video review.

Data. StatsBomb Open Data supplies matches, lineups, players, and event-level JSON covering **passes, shots, dribbles, carries, duels, and general events**. Foreign keys preserve the player-match-event relationship throughout the normalized database.

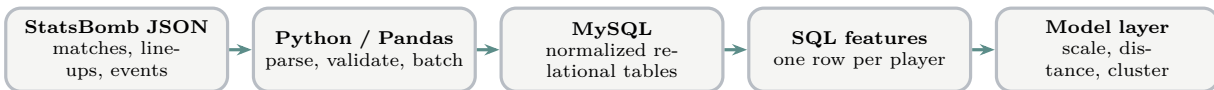


Figure 1. JSON → Python/Pandas → MySQL → SQL features → standardized player profiles.

Modeled dimensions. Passing volume, completion and progression; shooting volume, goals and xG; dribbling volume and success; carrying volume and progression. A completed pass has no recorded failure outcome, and an action is progressive when end x minus start $x \geq 10$. Players require events in at least five matches.

2. Methods: comparable rates, distance, and clustering

Step 1: normalize usage

Current baseline:

$$x_{pm} = \frac{\text{actions}}{\text{matches played}}$$

Minutes-aware extension:

$$x_{90} = \frac{\text{actions}}{\text{minutes}} \times 90$$

Step 2: standardize

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

Each feature receives mean 0 and standard deviation 1, making heterogeneous statistics comparable.

Step 3: compare

$$d(a, b) = \sqrt{\sum_j (z_{aj} - z_{bj})^2}$$

Smaller Euclidean distance indicates a more similar statistical style across the selected features.

K-Means archetypes. For each candidate K from 2 through 12, K-Means minimizes within-cluster squared distance. The repository uses a fixed random seed (42), ten initializations, and selects $K = 6$ from the bend in the inertia curve.

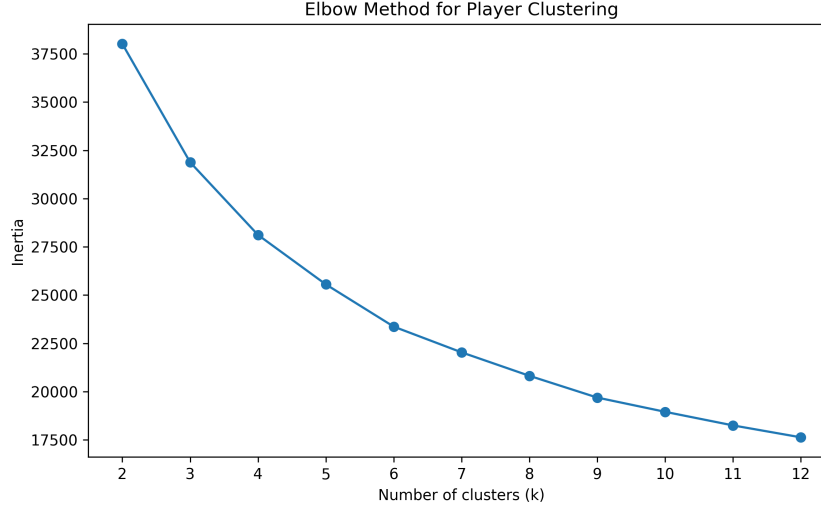


Figure 2. Model-selection diagnostic committed with the project. Inertia falls sharply at small K and begins flattening around six clusters, supporting a compact but differentiated archetype set.

Per-match versus per-90. The committed baseline uses per-match metrics because reliable minutes were not yet incorporated. Per-90 is the stronger next version: it corrects for substitute appearances and unequal playing time. Until then, match-rate comparisons should be treated as directional, particularly for players with limited minutes.

What is intentionally excluded from the feature vector? Player ID and name are identifiers only. Official position is also excluded, allowing the clusters to reveal performance roles rather than simply reproduce roster labels. Duels are parsed and available for exploration, but defensive duel, pressure, interception, and tackle features are not yet included in the clustering baseline.

3. Results: six interpretable playing archetypes

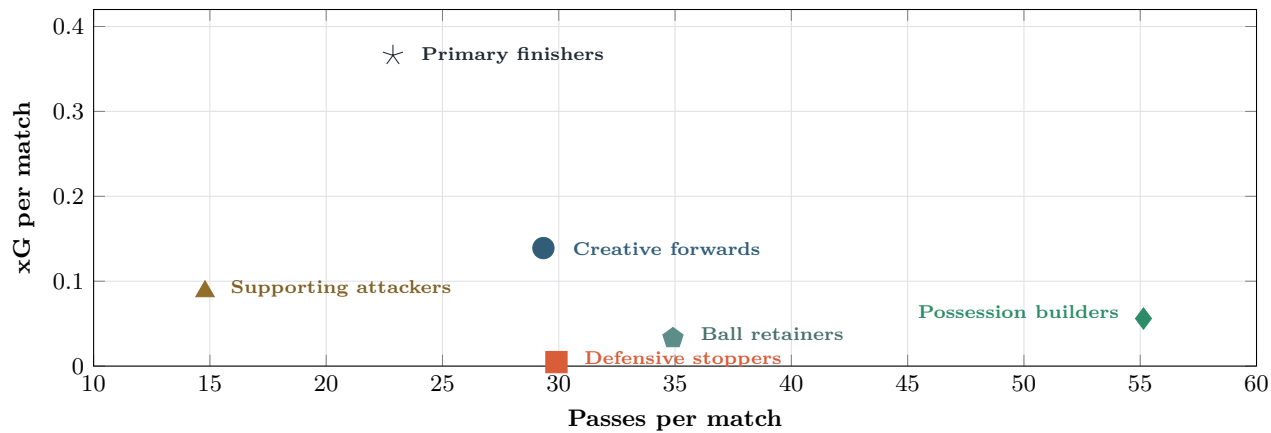


Figure 3. Cluster centroids reveal the principal tradeoff between possession involvement and goal threat. Values are means from the committed `cluster_profiles.csv`.

ID	Archetype	Statistical signature	Representative players documented in analysis
0	Advanced / creative forwards	Highest dribble volume; strong shooting and carrying	Lionel Messi, Neymar Jr., Johan Cruyff
1	Defensive stoppers	High progressive-pass share; almost no shooting or dribbling	Presnel Kimpembe, Jan Vertonghen, Vincent Kompany
2	Supporting attackers	Low passing volume; moderate shooting and dribbling	Mixed supporting attackers and wide players
3	Possession builders	Highest passing, progressive passing, and carrying volume	Marco Verratti, Kyle Walker, John Stones
4	Defensive ball retainers	Safe, high-volume circulation with limited attacking output	Ben Davies, Pablo Zabaleta, Oriol Romeu
5	Primary finishers	Highest shots, goals, and xG per match	Sergio Agüero, Romelu Lukaku, Edinson Cavani

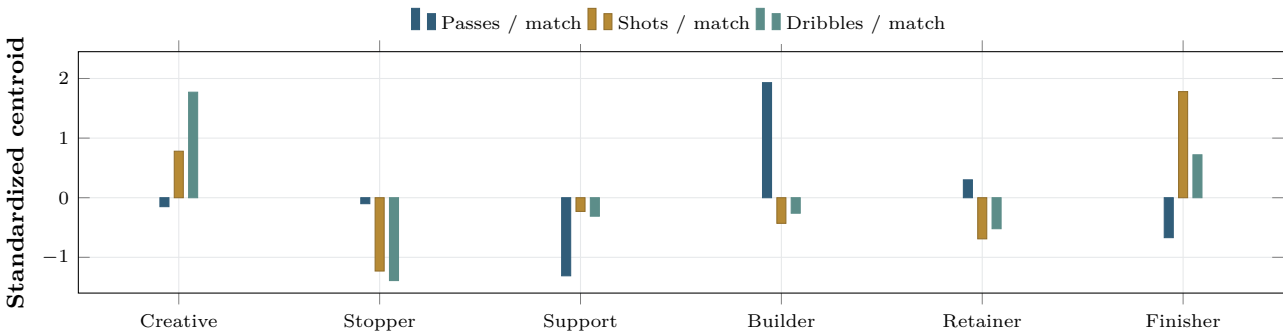


Figure 4. Standardized centroid comparison. Builders separate on circulation; finishers on shots; creative forwards on dribbles. Standardization is across the six cluster centroids for presentation.

4. From model output to a scouting decision

Example recommendations. The project implements nearest-neighbor ranking in the full standardized 13-feature space. The committed narrative report preserves representative members of each archetype but not the row-level distance export, so the examples below are presented as *style-consistent shortlists*, not fabricated distance rankings.

Scouting brief	Reference profile	Example shortlist	Why the model connects them
Creative, ball-dominant attacker	Lionel Messi	Neymar Jr.; Johan Cruyff	High dribble involvement, above-average shooting, and meaningful build-up participation
Penalty-area scoring threat	Sergio Agüero	Romelu Lukaku; Edinson Cavani; Pierre-Emerick Aubameyang	High shot, xG, and goal rates paired with lower passing involvement
Deep progression and circulation	John Stones	Kyle Walker; Marco Verratti	High pass and carry volume with strong progressive involvement and low shooting output
Defensive possession retention	Oriol Romeu	Ben Davies; Pablo Zabaleta	High-volume, conservative circulation with modest attacking activity

How a scout would use the findings

1. **Translate the brief.** Select the metrics that describe the tactical need: progression, retention, creation, or finishing.
2. **Choose a reference.** Use an incumbent or known target as the desired statistical profile.
3. **Generate a shortlist.** Retrieve low-distance players, preferably inside the same broad archetype.
4. **Apply constraints.** Filter by position, age, league, minutes, foot, fee, availability, and recruitment geography.
5. **Validate context.** Compare team possession, competition strength, tactical role, and video before recommending action.

Best use case. Reduce a very large player pool to a defensible set of stylistic candidates. The system is strongest as an early-stage discovery and comparison layer.

Do not use it alone to: Declare one player “better,” estimate transfer value, or infer off-ball defensive quality. Similarity means comparable measured style, not equal quality or suitability.

Limitations and the next analytical release

Current limitation	Why it matters	Recommended improvement
Position differences	A full-back and midfielder can be statistically close but tactically non-substitutable	Position-aware search or user-controlled position filters
Matches, not minutes	Starts and substitute appearances receive equal denominators	Derive reliable minutes and rebuild all volume metrics per 90
Missing defensive features	Tackles, pressures, interceptions, and richer duel context are absent from the model	Add on- and off-ball defensive rates and possession-adjusted measures
Mixed competitions / seasons	Team strength, style, and league quality influence observed events	Add season filters and competition-strength adjustment
Euclidean equal weighting	Every standardized feature contributes equally, regardless of the scouting brief	Learn or expose role-specific weights; evaluate cosine or Mahalanobis distance
No formal cluster validation	The elbow is interpretive and cluster labels are manually assigned	Add silhouette stability, PCA/UMAP views, and out-of-sample checks

Bottom line. Event data can make playing style searchable. This transparent baseline becomes recruitment-ready with minutes-aware normalization and stronger defensive context.

Source: Rahmah Abdulkarim, Football Player Similarity Analysis; data: StatsBomb Open Data.